

A. Descriptive Statistics Report

本研究共纳入 40 份完整问卷作为后续分析样本。参与者整体以年轻大学生为主，其中 18-24 岁占 95.0% (n = 38)，25-34 岁占 5.0% (n = 2)。在性别分布上，男性参与者占 62.5% (n = 25)，女性占 30.0% (n = 12)，非二元性别占 5.0% (n = 2)，另有 2.5% (n = 1) 选择不愿透露。

从专业背景来看，样本主要来自 STEM 相关专业，占 95.0% (n = 38)，另有 5.0% (n = 2) 来自商科背景。就学习阶段而言，本科生占 87.5% (n = 35)，研究生占 12.5% (n = 5)。此外，所有参与者均已完成或正在修读编程相关课程，其中 77.5% (n = 31) 正在修读相关课程，22.5% (n = 9) 已完成相关课程。这表明样本整体具有较明确的编程学习背景，适合用于分析学生在编程学习中使用生成式 AI 工具的经验、态度与需求。

在编程语言经验方面，参与者最常掌握的语言为 Python 和 Java，二者均占 90.0% (n = 36)。此外，62.5% (n = 25) 的参与者掌握 C++，57.5% (n = 23) 掌握 HTML，52.5% (n = 21) 掌握 JavaScript/TypeScript，52.5% (n = 21) 掌握 C，45.0% (n = 18) 掌握 CSS，20.0% (n = 8) 掌握 C#。另有 10.0% (n = 4) 选择了其他编程语言，并补充了 Matlab、R、Go、Dart、Swift、Nix、Haskell 等语言。这说明参与者不仅具有基础编程经验，也覆盖了多种常见编程语言。

在 AI 工具使用经验方面，绝大多数参与者曾使用通用型 AI 助手进行编程学习或编程任务，占 97.5% (n = 39)；同时，80.0% (n = 32) 的参与者使用过 coding agent 或专门面向编程的 AI 工具。仅 2.5% (n = 1) 报告使用了其他类型的 AI 工具。关于 AI 编程工具的使用时长，55.0% (n = 22) 的参与者已使用超过一年，15.0% (n = 6) 使用 6-12 个月，15.0% (n = 6) 使用 3-6 个月，10.0% (n = 4) 使用 1-3 个月，5.0% (n = 2) 使用不足一个月。关于使用频率，42.5% (n = 17) 每天使用 AI 工具进行编程，40.0% (n = 16) 每周使用 2-6 次，另有少数参与者报告每周一次、每月一次、每 2-3 个月一次或低于每月一次的使用频率。

总体而言，该样本由具有编程学习经验且普遍接触过 AI 编程工具的学生构成。多数参与者来自 STEM 背景，掌握 Python、Java 等常见编程语言，并且在生成式 AI 编程工具方面具有较高的使用频率和较长的使用经验。因此，该样本能够为理解学生在 AI 辅助编程学习中的实际使用方式、依赖风险、感知收益以及对负责任 AI 编程学习支持的需求提供有针对性的形成性证据。

The formative study analyzed 40 complete questionnaire responses. The sample was primarily composed of young university students: 95.0% of participants were aged 18-24 (n = 38), and 5.0% were aged 25-34 (n = 2). In terms of gender, 62.5% identified as male (n = 25), 30.0% as female (n = 12), 5.0% as non-binary (n = 2), and 2.5% preferred not to disclose their gender (n = 1).

Participants were predominantly from STEM-related majors (95.0%, $n = 38$), with a small proportion from business-related backgrounds (5.0%, $n = 2$). Most participants were undergraduate students (87.5%, $n = 35$), while 12.5% were graduate students ($n = 5$). All participants had either completed or were currently taking programming-related courses: 77.5% were currently enrolled in such courses ($n = 31$), and 22.5% had previously completed them ($n = 9$). This indicates that the sample had a clear programming-learning background and was suitable for investigating students' experiences, perceptions, and needs regarding generative AI tools in programming learning.

Participants reported familiarity with multiple programming languages. Python and Java were the most common, each reported by 90.0% of participants ($n = 36$). Other frequently reported languages included C++ (62.5%, $n = 25$), HTML (57.5%, $n = 23$), JavaScript/TypeScript (52.5%, $n = 21$), C (52.5%, $n = 21$), CSS (45.0%, $n = 18$), and C# (20.0%, $n = 8$). In addition, 10.0% of participants selected other programming languages ($n = 4$), including Matlab, R, Go, Dart, Swift, Nix, and Haskell. These results suggest that the participants had broad exposure to common programming languages and development contexts.

The sample also showed substantial experience with AI tools for programming. Nearly all participants had used general-purpose AI assistants for programming-related tasks (97.5%, $n = 39$), and 80.0% had used coding agents or programming-specific AI tools ($n = 32$). Regarding duration of AI tool use for programming, 55.0% had used such tools for more than one year ($n = 22$), 15.0% for 6-12 months ($n = 6$), 15.0% for 3-6 months ($n = 6$), 10.0% for 1-3 months ($n = 4$), and 5.0% for less than one month ($n = 2$). In terms of frequency, 42.5% reported daily use ($n = 17$), and 40.0% reported using AI tools for programming two to six times per week ($n = 16$). Smaller groups reported weekly, monthly, every two to three months, or less-than-monthly use.

Overall, the sample consisted of students with relevant programming-learning experience and frequent exposure to AI programming tools. Most participants came from STEM backgrounds, were familiar with common programming languages such as Python and Java, and had substantial experience using generative AI tools for programming. These characteristics make the sample appropriate for a formative investigation of how students use AI in programming learning, what benefits and risks they perceive, and what forms of responsible AI-supported programming assistance they may need.